Yianni Karakoulas

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Education

University of Toronto — Toronto, ON
Bachelors of Computer Science, Honours Bachelors of Science — *Expected May 2030*

Upper Canada College — Toronto, ON
International Baccalaureate (IB) Diploma — *September 2018 - May 2026*

Projects

Pathology Foundation-Model Fusion | *Python, PyTorch, Hugging Face, scikit-learn, W&B* — October 2026

- Built a weakly supervised pathology pipeline that processed 368,250 tiles from 203 whole-slide images through UNI2-h, Virchow2, and Phikon-v2, using attention-based multiple-instance learning to classify seven lung adenocarcinoma growth patterns.
- Evaluated agent-selected fusion strategies using patient-level splits, five-fold cross-validation, equal-budget controls, and bootstrap uncertainty estimates, achieving 0.874 macro-AUROC on a one-time test set of 41 patients.

Your Trusted Circle (YTC) | *TypeScript, Node.js, Hono, MongoDB, Railway* — June 2026 – September 2026

- Designed and deployed a Hono/MongoDB REST API for authenticated group-safety sessions, moving check-in deadlines and idempotent alert escalation server-side so incidents progressed even when clients disconnected.
- Delivered 12 merged PRs in 36 hours, implementing consent-checked ephemeral location endpoints, atomic alert cancellation, hash-chained incident timelines, and voice-agent and Incident Card APIs; validated by a 169-test Vitest suite.

Memory Influence Debugger | *Next.js, TypeScript, React, Backboard API, Vitest* — July 2026 – September 2026

- Built and deployed a Next.js debugger that compares memory-conditioned and memory-disabled plans across three action schemas, surfaces field-level changes, and identifies likely memory influence before execution.
- Redesigned memory-relevance and polarity handling after a 40-scenario benchmark exposed retrieval failures, raising regression correctness from 2.5% to 95.0% and eliminating 19 false positives.

Awards

International Climate Science Olympiad : 1st/37,500 Contestants, $3500 USD Prize and sponsored trip to COP29

University of Western Ontario National Scholarship, Finalist : Top 100/1500+ Nationally

Experience

Mathematics Tutor — August 2023 - June 2026
Spirit of Math Schools — *Toronto, ON*

- Instructed advanced mathematics topics in linear algebra and calculus to classes of 20 high-school students twice weekly
- Diagnosed errors in student work and delivered targeted feedback, reinforcing mathematical reasoning, accuracy, and independent problem-solving.

Director of Programs and Events — June 2025 – March 2026
World Affairs Conference — *Toronto, ON*

- Initiated the Service Fair, showcasing student led organizations and initiatives in philanthropy, leading the application and interview process
- Led program and event operations for a conference serving 1000+ delegates across 50+ schools, coordinating schedules, speakers, logistics, and student teams at the biggest current affairs conference in North America

Technical Skills

Languages: Python, TypeScript
Frameworks: Next.js, React, TensorFlow, Keras
Developer Tools: Git, GitHub, npm, Vercel, Google Colab, Weights & Biases (wandb)
Certifications: NIDA Good Clinical Practice (GCP), Ethical Conduct for Research Involving Humans (TCPS 2: CORE 2022)